
Index

Introduction to Environment	1
Man & Environment Relationship	5
Environmental Science/Education	7
Environmental Degradation.....	9
Role of Engineering Principals.....	11
Impact of Technology on the Environment	13

Introduction to Environment

Introduction

- The word Environment is derived from the French word “Environner”, meaning to encircle or surround.
- All the biological and non-biological things surrounding an organism are included in the environment.
- Environment can be defined as “**External surroundings and conditions which directly or indirectly affect the living organisms.**”
- “Environment is a total of water, air, and land, the interrelationship among themselves and also with human beings, other living organisms and property.”
- Environmental science is an interdisciplinary academic field that integrates physical and biological sciences, (including but not limited to Ecology, Physics, Chemistry, Biology, Soil Science, Geology, Atmospheric Science, and Geography) to the study of the environment, and the solution of environmental problems.
- Environmental science provides an integrated, quantitative, and interdisciplinary approach to the study of environmental systems. Related areas of study include environmental studies and environmental engineering. Environmental studies incorporate more of the social sciences for understanding human relationships, perceptions, and policies toward the environment.
- Environmental engineering focuses on design and technology for improving environmental quality. Environmental scientists work on subjects like the understanding of earth processes, evaluating alternative energy systems, pollution control and mitigation, natural resource management, and the effects of global climate change. Environmental issues almost always include an interaction of physical, chemical, and biological processes.

The environment consists of 2 entities:

- Biotic entities
- Abiotic entities
- Biotic environment is made up of all living organisms (Plants, animals & microorganisms) including their reactions, interactions, and interrelated actions. These biotic components are in a dynamic state, they constantly affect each other and cannot be isolated from each other.
- Abiotic is composed of external physical factors like temperature, humidity, water, soil, minerals, gases, etc. It provides both habitation and raw materials for the synthesis of organic food.

Unit 1 – Multidisciplinary Nature of Environment

→ Any constituents of the environment which directly or indirectly affect the growth and development of an organism are called an environmental or ecological factor. Examples of environmental factors are climatic factors, topographic factors, biotic factors, and fire.

Organic & Inorganic

→ Organic substances are made up of C & H (carbon and hydrogen), whereas inorganic substances lack carbon or hydrogen or both in their chemical formula.

→ Organic substances: methane, protein, lipid, fat, propane, etc.

→ Inorganic substances: salt, washing powder, CO₂, Oxygen, etc.

Components of Environment

The environment can be divided into **four components**: Atmosphere Hydrosphere, Lithosphere, and Biosphere.

(1) **Atmosphere**: The atmosphere is the mixture of various gases and water vapor and subatomic particles that entirely covers the earth extending outward several thousand kilometers.

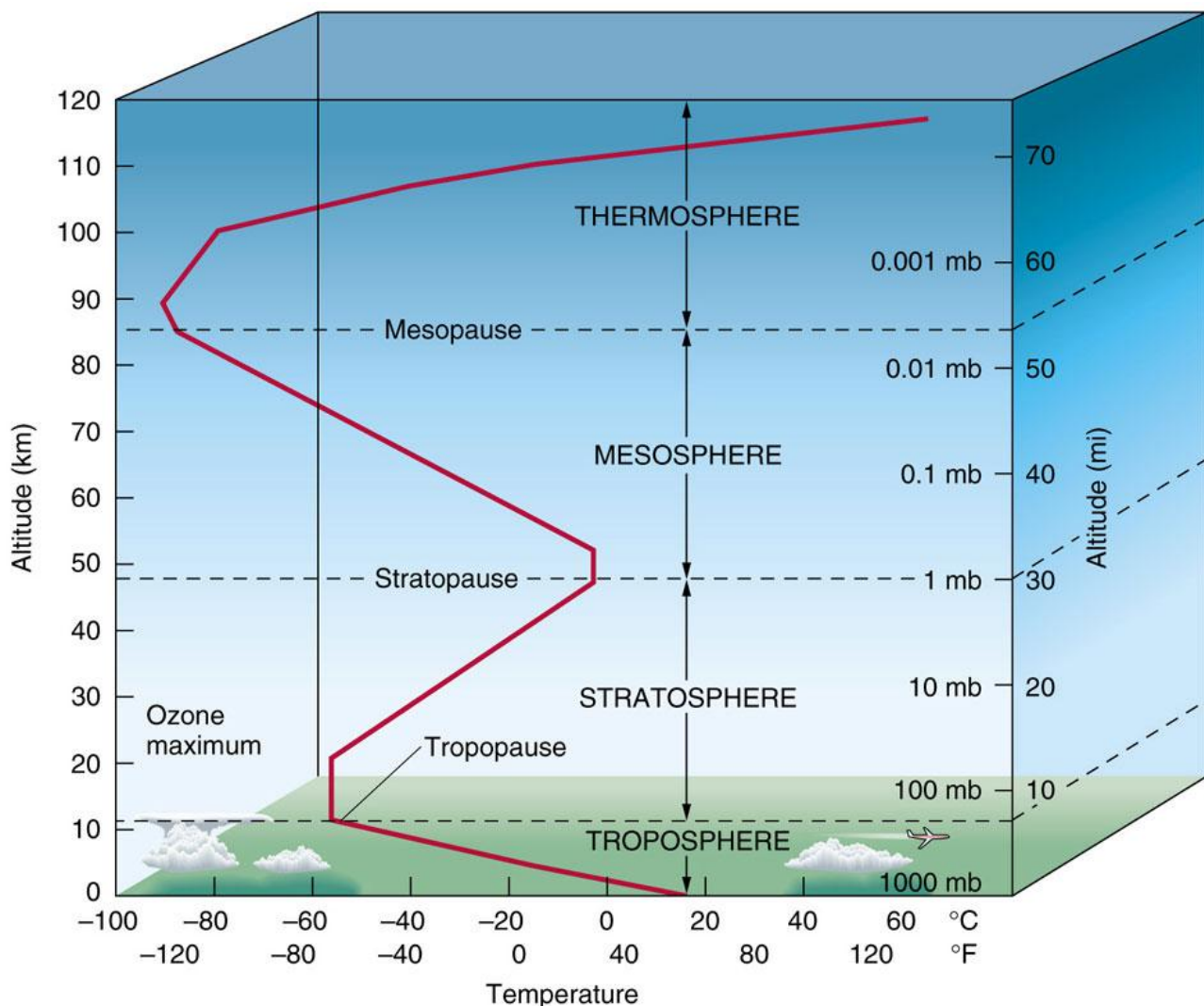
- The major gases in pollution-free dry air are nitrogen 78%, oxygen 21%, argon 1%, and carbon dioxide.
- The minor gases include Neon, Helium, Methane, Hydrogen, Carbon monoxide, Ozone, etc.

Based on the temperature profile and other related phenomena, the atmosphere is divided into four major layers.

- **Troposphere**; is the lower portion of the atmosphere which extends up to 8 Km at the poles and 16 Km at the equator on average it extends up to 10-12 km of the earth's surface. The temperature in this region decreases at the rate of 5-7 c/km there are strong vertical air movements in this region which are responsible for the rapid and complete mixing.
- **Stratosphere**; It is above the troposphere and extends up to 50-55 km up to about 20 Km temperature remains constant and then increases with an increase in height. The ozone layer is in this layer.
- **Mesosphere**; It is the layer above the stratosphere and extends up to 80 km. here temperature decreases slowly with altitude but then sharply to about -75 °C.
 - It is said to coldest region in atmosphere.
- **Thermosphere**; Thermosphere extends from 80 km to 550 km. Thermosphere is divided into two parts: the upper part and the lower part. The upper part is known as the

exosphere and the lower part is known as the ionosphere. The ionosphere helps in radio communication and exosphere is used as an orbit for satellites.

General temperature profile in atmosphere can be defined as **zigzag pattern**, as shown in the figure.



(2) **Hydrosphere:** The hydrosphere is the water environment. About 70-75% of the earth is covered by water. Water is available in seas, oceans, lakes, rivers, glaciers, etc.

- It is estimated that the hydrosphere contains about 1360 million cubic km of water.
- Saline water amounts to 97% available which is in oceans.
- Fresh water is about 3%, out of which 2.5% in groundwater, glaciers and ice caps while the remaining 0.5% of fresh water is available as rivers and lakes.
- Groundwater can be considered for largest storage of freshwater for human, as it is quite accessible,

- Water is the main constituent in all living organisms and acts as an important resource for human life.

(3) Lithosphere: The outer soil crust of the earth is the lithosphere. The living organisms, plants, and vegetation are supported by the lithosphere.

- It also contains resources like minerals, organic as well inorganic matter, and to some extent air and water.
- Lithosphere plays an important role as it not only produces food for human beings and animals but also the decomposition of organic wastes is carried out by a host of microorganisms in the soil.

(4) Biosphere: The biosphere is the thin outer crust of the earth which included all the living organisms and their environment. For Example:

- It extends from the lowest sea bed level to about 24 km of the atmosphere. Thus, the biosphere consists of lithosphere, atmosphere, and hydrosphere."
- The living organisms interact with one another in the biosphere and sustain their life. The life-supporting resources are also available from the biosphere.
- The waste products in gaseous, liquid, and solid waste forms are discharged into the biosphere. Though the sustaining and assimilative capacity of the biosphere is tremendous it is not infinite.

Man & Environment Relationship

Man & Environment Relationship

- The two words Man and environment are not new to human history and the interrelation between them is well established.
- Thinking about the environment is as old as our first human ancestors. Their survival depended on knowledge of it. Concern for the environment is also not new.
- Since ancient times, people have known the importance of preserving it. Worshipping of trees and rivers, animals and birds were not based on superstition; but there was a hidden message about preserving and protecting the environment.
- The religious rituals served an important purpose - they made people aware of the environment and its importance and so indirectly helped preserve and nourish it.
- Religion is used largely to ensure community participation in all those practices that would lead to the maintenance of ecological balance.
- Man is the only living organism capable of modifying its surrounding environment according to the need. Other animals change according to the environment.
- From the time of evolution many changes are recorded in human history, which broadly includes, increase in resource usage, increase in convenience & quality of life, reduction in human effort.
 - First man started hunting animals and cutting trees for his basic need gradually with time he started cultivating food grains for that he started hunting animals and cutting trees for his basic needs.
 - Gradually with time he started cutting forests and converting them into grasslands/agricultural fields.
 - To increase his comfort, he started disturbing every component of the environment.
 - With the start of the industrial revolution the total scenario changed.
 - Everything changed, the use of fossil fuel for the generation of power for running vehicles, and many industries started to produce the product which increase pollution of all types.
 - Quantity of carbon dioxide emission increased tremendously which has started showing effect in the form of discharge of industrial waste and sewage.
 - Major environmental issues arising due to human activities are global warming, acid rain, ozone depletion, and population explosion.

- Emission of CO₂ and the other gases in the atmosphere from fossil fuel burning and other human activities may raise the temperature of the earth's lower atmosphere by several degrees by 2050. This would disrupt food production and flooding of low-lying coastal cities and croplands.
- An estimated 36500 species of plants and animals become extinct each year, mostly because of human activities. The main factors which affect the distribution of population and human settlement are the Relief of land, climate, soils, mineral deposits, and water supply.

Environmental Science/Education

- Environmental science in its broadest sense encompasses all the fields of natural science. It is interdisciplinary.
- The study by environment scientists has been focused on the natural environment i.e. the atmosphere, the land, the water, and the inhabitants as differentiated from the built environment.
- Natural science- includes such diverse disciplines as biology, chemistry, geology, physics, and environmental science.
- It refers to the study of basic principles & concepts of natural sciences and the application thereof to understand the multidisciplinary and complex nature of the environment and its related problems.

Scope of Environmental Studies:

- Environment is not limited to issues of sanitation and health but it is now concerned with pollution control, biodiversity conservation, waste management, and conservation of natural resources.
- The following area requires knowledge of environmental studies and has good opportunities.
 - Natural Resource Conservation
 - Environmental Pollution Control
 - Environmental management
 - Social Development
 - Research and development
 - Ecosystem Structure and Function

Importance of environmental education:

- It gives us a basic understanding of various environmental aspects and associated problems.
- It teaches us the concept of sustainable development.
- It gives an idea about the beneficial use of natural resources without damaging them much.
- It teaches us how to conserve energy and save our planet.
- It develops skills to identify environmental problems and their solution.

The objective of Environmental Studies:

- Increase awareness of the total environment.

Unit 1 – Multidisciplinary Nature of Environment

- Increase the knowledge of the environment.
- Improve attitude towards the environment.
- Motivate environmental protection.

Environmental Degradation

→ It is the direct and indirect deterioration of the environmental quality due to different activities of living beings that pollute key elements like air, water, and soil.

→ Environmental degradation can occur naturally or through human activities.

Natural factors:

- Natural factors causing environmental degradation are draughts, storms and floods, earthquakes, tsunamis, and volcanic eruptions.
- These factors not only cause physical and agricultural damage but are also responsible for disruption of property and essential services like electricity, and water supply.

Manmade factors:

- These human factors include deforestation, industrialization, and urbanization. The environment is degraded because of the following reasons:
- High growth of the human population and poverty is mainly responsible for environmental degradation.
- Unplanned and heavy consumption of natural resources in which even renewable resources are not given sufficient time to renew.
- Ecological imbalance created by human and animal activities.
- Overuse of resources like water, forest, land, and energy which leads to solid, liquid, and air pollution.
- Heavy industrialization leads to air and water pollution which causes respiratory infections, toxic chemicals and heavy metals, contamination of water, use of pesticides, and release of a high amount of greenhouse gases contributing to the cause of cancer, birth defects, low immune power, global warming, climatic changes, and other health hazards.

Unit 1 – Multidisciplinary Nature of Environment

- Alarming rate (speed) of population growth has put excessive pressure on all of the natural resources.
- Total environmental degradation & pollution, arise mainly due
 - Consumption of natural resources by overpopulation of developing countries
 - Wasteful overconsumption of natural resources in developed countries.
- In developing countries 3 Ps are key factors for rapid environmental degradation.
 - Population
Higher demand leading to high resource usage and pollution.
 - Poverty
Unemployment
Social issues relating to environment
Lack of fulfillment of basic needs
 - Pollution

Role of Engineering Principals

Role of Engineers:

- Environmental science is directly connected to various modes of life. With technological advances, the application of different engineering solution principles in our daily lives is increasing.
- Every engineering solution will consume some sort of resource.
- Resources can be either;
 1. Exhaustible: resources which are present in limited amount
 - For example, Fossil fuel, petrol, crude oil, coal, etc.
 2. Inexhaustible resources which are present in unlimited amount
 - For example, solar energy, wind energy, etc.
- Developing technologies that eliminate the use of environmentally dangerous compounds, be the focus in engineering disciplines to address environmental issues.
- **Mechanical Engineering:** Mechanical engineering deals with the design and operation of many combustion processes.
 - Any combustion process will require fuel. If a mechanical engineer is aware of the impact of the use of different types of fuels on the environment, he would try to develop a process that uses greener fuels and thus reduce pollution.
 - He/she would also try to increase the efficiency of combustion Processes, as reduce pollution, because incomplete combustion, leads to the production of carbon monoxide and other air pollutants.
 - If they are aware of the positive impacts of renewable energy sources through the study of environmental science, they would try to develop new technologies to harvest wind energy, hydropower, etc. Mechanical engineering also deals with the subject of refrigeration.
 - Use of refrigerants that have higher global warming potential and which cause depletion of the ozone layer should be discouraged and the focused should be on developing technologies that may eliminate the use of these environmentally dangerous compounds.
- **Civil Engineering:**
 - Demolition and construction waste is produced in large quantities in various works related to civil engineering.

- If a civil engineer is aware of the problems caused by handling and disposing of solid waste, he/she would try to reuse this waste as a construction material, the benefit of which will be two-fold,
 1. Reduction of usage of new resources
 2. Reduction in the quantity of waste that has to be disposed of properly to minimize the risk to the environment.
- Construction/ demolition waste can be used to replace aggregates in concrete. They can also develop methods/ technologies which could be used to use plastic: a very dangerous pollutant, for bituminous road construction.

→ **Computer/IT Engineering:** Environmental science includes the study of e-waste; its sources and its impact on the environment.

- Computer engineers should try to develop technologies that, can be used to upgrade existing computers and other IT gadgets easily, for reducing the need to replace the existing ones.
- Computer engineers can promote digitalization which is digital storage and management of documents or data, which helps in minimizing physical storage spaces required.
- Some of the characteristics of digitalization are More security concerns, less land use for storage, greater transparency among users, more possibilities of theft.
- Thus, there would be a reduction in the generation rate of e-waste and problem related to disposal. Similarly, knowledge of environmental science will also help other engineering disciplines achieve a border goal of sustainable development.

→ **Electrical Engineering:** Production of electrical energy by conventional sources causes depletion of natural resources and large amounts of pollution.

- Power sector is one of the largest contributors of greenhouse gases.
- Reduction in power consumption can reduce the emission of greenhouse gases and air pollution to a great extent.
- Basic knowledge of different types of pollutants produced in power generation and their impacts on the environment will inspire electrical engineers to, develop energy-efficient devices/equipment which will **reduce power consumption**, which in turn will lead to resource conservation and pollution abatement.

Impact of Technology on the Environment

Impacts of Technology on the Environment:

- The technology is developed and used for human well-being.
- Technology is developed by utilizing various resources available.
- The goal of utilizing natural resources according to environmental science should be, to use them beneficially without causing extensive harm
- The intelligent application of technology results in human well-being but not without causing environmental disruptions which undermine well-being.
- Technology has played a major role in the well-being of humans but it has some negatives as well.
- The scale and the rapid rate with which technology is being developed and used make all the difference.
- No technology can be completely free of environmental impacts. This statement simply says that the development and use of any technology lead to environmental costs exceeding its benefits.
- The direct effects include accidents, the release of pollutants affecting human health, exhaustion of resources, changes in landscapes, etc.
- The indirect effects interfere with the vital services supplied to humanity by natural ecological systems e.g., life expectancy, control of pollution through natural processes.
 - Life expectancy is the prime example of indirect effect.

Impact of Urbanization

- Urbanization means conversion of rural area into urban area.
- Increasing the facility available in the area can be considered as development activity, which is synonymous to urbanization.
- Urbanization has some major effects on environment as stated below;

Impact on atmosphere

- Due to urbanization, there is an increase observed in the CO₂ and proportionate decrease in the amount of O₂.
- Surrounding temperature may rise due to higher human interaction or industrial activities.
- Increase in air pollution originating from exhaust of vehicles and various modes of transport available.
- Presence of particulate matter (small liquid and solid particles, i.e., dust smoke, dew) and noxious fume (poisonous or highly harmful smoke and gases, chemical gases) increases.

Impact on hydrosphere,

- Due population and various industry and commercial activity there is **higher demand** of water resources, and problems like resource scarcity may also arise.
- There is increase in pollution of water resources because of high ground usage, waste disposal.
- Drainage patterns affected on and below ground level is affected because of various mode of transportation.
- Further, rain and surface water pollution and results in high wastewater generation.
- Urban areas have high amount of waste present on the ground, leaching of this waste may reach ground water reserves and causes pollution and unsatisfactory water quality.

Impact on lithosphere,

- Increased land use for residing (living purpose) and decrease in agriculture land
- Natural land features are affected.
- Increase in mode of transport, vehicular density, and air pollutants.
- Due to roadways and railway construction, there is disruption /disfigurement of the landscape

Unit 1 – Multidisciplinary Nature of Environment

→ Urban waste disposal disturbs lithosphere quality

Environmental Component	Population	Land use	Transportation	Services
Atmosphere	Increased release of CO ₂ , Decreased O ₂ production	Increased average temperature	Air pollution from combustion of fuels	Particulate matter and noxious fumes
Hydrosphere	Greater demand of water resources	Increased pollution in most of the resources	Rain and surface water polluted with lead, drainage pattern change	Leaching of pollutants from landfills reaches ground water reserves
Lithosphere	Increased land use for residing and decrease in agriculture land	Complete changes due to construction and landscape.	Disruption or disfigurement of the landscape	Urban waste disposal disturbs the landscape